

# Cucumber BDD with Java & Selenium

## POM.XML

```
<!-- selenium-java -->
<dependency>
    <groupId>org.seleniumhq.selenium</groupId>
    <artifactId>selenium-java</artifactId>
    <version>4.34.0</version>
</dependency>
<!-- cucumber-java -->
<dependency>
    <groupId>io.cucumber</groupId>
    <artifactId>cucumber-java</artifactId>
    <version>7.27.0</version>
</dependency>
<!-- cucumber-testng -->
<dependency>
    <groupId>io.cucumber</groupId>
    <artifactId>cucumber-testng</artifactId>
    <version>7.27.0</version>
</dependency>
```

## Scenario1.feature

Feature: Login Functionality

Scenario Outline: Successful Login with valid credentials

Given user is on login page

When user enters "<username1>" and "<password1>"

Then user should see "<result>"

Examples:

```
|username1|password1|result|
|Admin|admin123|Dashboard|
|invalid|wrongpassword|Invalid credentials|
```

## LoginSteps.java

```
package com.testing_apps.testing_app1.stepDefinations;

import java.time.Duration;

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.support.ui.ExpectedConditions;
import org.openqa.selenium.support.ui.WebDriverWait;
import org.testng.Assert;

import io.cucumber.java.en.Given;
import io.cucumber.java.en.Then;
import io.cucumber.java.en.When;

public class LoginSteps {
    private WebDriver driver;

    @Given("user is on login page")
    public void user_is_on_login_page() {
```

# Cucumber BDD with Java & Selenium

```
// Write code here that turns the phrase above into concrete actions
driver = new ChromeDriver();
driver.manage().window().maximize();
driver.get("https://opensource-
demo.orangehrmlive.com/web/index.php/auth/login");
}

@When("user enters {string} and {string}")
public void user_enters_and(String username1, String password1) {
    // Write code here that turns the phrase above into concrete actions
    WebDriverWait wait = new WebDriverWait(driver,
Duration.ofSeconds(30));

    wait.until(ExpectedConditions.visibilityOfElementLocated(By.xpath("//input
[@name='username']")))
        .sendKeys(username1);

    wait.until(ExpectedConditions.visibilityOfElementLocated(By.xpath("//input
[@name='password']")))
        .sendKeys(password1);

    wait.until(ExpectedConditions.elementToBeClickable(By.cssSelector("button[
type='submit']"))).click();
}

@Then("user should see {string}")
public void user_should_see(String expectedText) {
    System.out.println(expectedText);
    // Wait for either Dashboard or Invalid credentials to appear
    WebDriverWait wait = new WebDriverWait(driver,
Duration.ofSeconds(10));

    try {
        // Case 1: Login successful → Dashboard visible
        WebElement dashboardHeader = wait

        .until(ExpectedConditions.visibilityOfElementLocated(By.xpath("//h6[text()
='Dashboard']")));
        String actualText = dashboardHeader.getText();
        System.out.println(expectedText + " --:-- " + actualText);
        Assert.assertEquals(actualText, expectedText, "Login successful,
but text mismatch!");
    } catch (Exception e) {
        // Case 2: Invalid credentials message visible
        WebElement errorMsg = wait.until(ExpectedConditions

        .visibilityOfElementLocated(By.xpath("//*[contains(text(),'Invalid
credentials')]")));
        String actualText = errorMsg.getText();
        System.out.println(expectedText + " --:-- " + actualText);
        Assert.assertEquals(actualText, expectedText, "Login failed, but
error message mismatch!");
    }
}
}
```

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## TestRunner.java

```
package com.testing_apps.testing_appl;

import io.cucumber.testng.AbstractTestNGCucumberTests;
import io.cucumber.testng.CucumberOptions;

@CucumberOptions(features = "src/test/resources/features", glue = {
    "com.testing_apps.testing_appl.stepDefinations" }, plugin = {
    "pretty",
    "html:target/cucumber-report.html" }, monochrome = true)
public class TestRunner extends AbstractTestNGCucumberTests {

}
```